VIP Interneurons and Predictive Coding in Mouse Visual Cortex

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Pod: Glamorous Bleeding Heart

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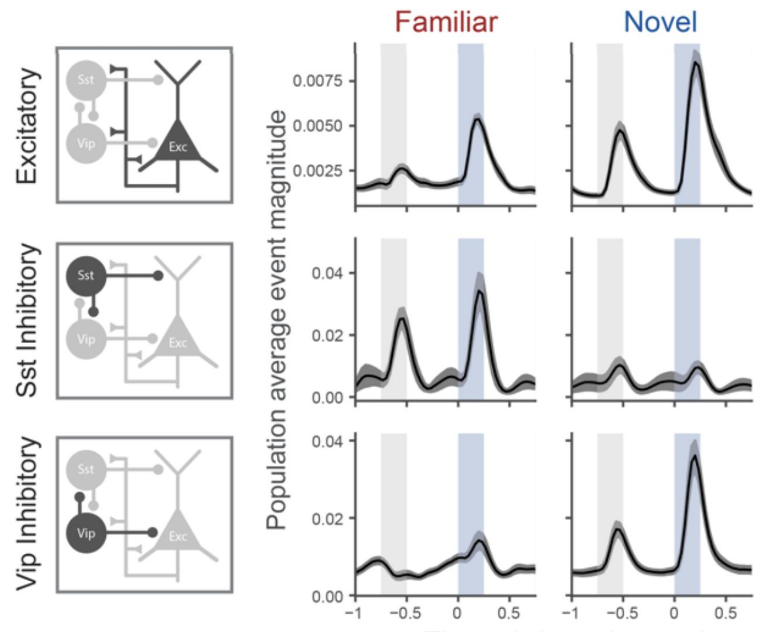
Introduction: Predictive Coding

Predictive coding helps the brain process new information by continuously updating internal models. Neurons encoding prediction errors show higher activity for novel stimuli, enabling efficient novelty detection.

VIP interneurons in deeper cortical layers suppress SST inhibitory neurons, which target novelty-encoding excitatory neurons in superficial layers. This mechanism supports the brain's ability to maintain the activity of neurons involved in encoding novel stimuli.



Garrett et al., 2023

Hypothesis

Vasoactive intestinal peptide (VIP) interneurons in the deeper cortical layers play a significant role in modulating predictive coding.

1. we propose that these VIP interneurons suppress somatostatin (SST) inhibitory feedback, which targets novelty-encoding excitatory neurons in the shallower cortical layers.

2. This suppression is expected to maintain the activity of excitatory neurons involved in encoding novel stimuli, thereby facilitating their contribution to the generation and updating of predictive models in response to new information.

We anticipate observing activation of VIP interneurons in the absence of activity from familiarity-encoding excitatory populations in deeper layers, followed by subsequent activation of novelty-encoding excitatory neurons in shallower layers. This would suggest a feedback mechanism modulating predictive coding functions in cortical regions near the retina.

Dataset

The Allen Institute Dataset is composed of neural data taken from mice

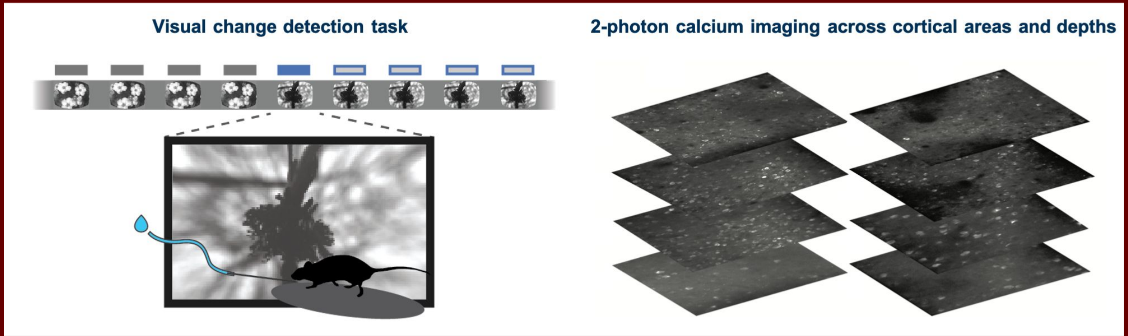
Shows differences between novel and familiar images for a mouse

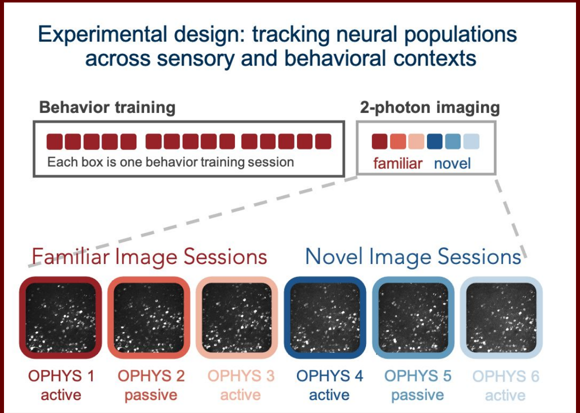
Contains lots of points that distinguishes individual neurons and depths

The dataset includes various depths and layers of the brain

Providing a comprehensive view that allows for detailed investigation

Essential for accurately understanding the functional and structural connectivity within the brain





Covariance Matrix

Here is a covariance matrix showing the relationship between the different depth layer's dff outputs.